DORIAN W. P. AMARAL

Institut de Física d'Altes Energies (IFAE),
The Barcelona Institute of Science and Technology,
Campus UAB, 08193 Bellaterra (Barcelona), Spain

damaral@ifae.es   dwpamaral.github.io

0000-0002-1414-932X   D.W.P.Amaral.1   dwpamaral

EDUCATION

2018–2022 **PhD** Astroparticle Physics
Institute for Particle Physics Phenomenology (IPPP)
Durham University, UK
Title: *"A New Era for Direct Detection Experiments: Probing New Neutrino Physics at Dark Matter Detectors with Solar Neutrinos"*
Supervisor: David G. Cerdeño

2014–2018 **MPhys** Theoretical Physics (First Class Honours)
Durham University, UK
Dissertation Title: *"Novel Techniques for Dark Matter Direct Detection"*

ACADEMIC POSITIONS

2025–Present **Postdoctoral ERC Researcher, IFAE (Barcelona, Spain)**
Developing theoretical frameworks and phenomenology for high-frequency gravitational-wave detection using electromagnetic cavities with Diego Blas.

2022–2025 **Postdoctoral Associate, Rice University (Texas, USA)**
Led the first search for ultralight dark matter using a levitated sensor, worked on ultralight dark matter phenomenology with quantum sensors, and applied machine learning techniques to neutrino physics at direct detection experiments with Christopher Tunnell.

PUBLICATIONS

Preprints

1. *"Multi-messenger signal from QCD-Axion Ultracompact Minihalos"*, **D. W. P. Amaral**, E. D. Schiappacasse, and K. Tapia-Rebolledo, arXiv:2608.25338 [hep-ph]. Submitted to PRD.

2. *"First Search for Ultraheavy Dark Matter Using a Magnetically Levitated Particle"*, D. G. Uitenbroek, **D. W. P. Amaral**, J. Qin, J. Langendorff, A. Gingerich, T. H. Oosterkamp, and C. D. Tunnell, arXiv:2608.20464 [hep-ph]. Submitted to PRL.

3. *"SNuDD: Solar Neutrinos for Direct Detection"*, **D. W. P. Amaral**, D. Cerdeño, A. Cheek, V. Costa, and P. Foldenauer, arXiv:2607.22817 [hep-ph]. Submitted to JHEP.

4. *"A Levitated-Magnet Vector Force Sensor for Spin-Dependent Exotic Interactions"*, **D. W. P. Amaral**, L. Cong, T. M. Fuchs, H. Ulbricht, and D. Budker, arXiv:2606.30781 [hep-ph]. Submitted to PRL.

5. *"Global detector network to search for high-frequency gravitational waves (GravNet): conceptual design"*, **D. Amaral** et al., arXiv:2603.24645 [astro-ph.IM]. Submitted to PRD (Accepted).

Peer-reviewed Publications

6. *"Model-independent searches of new physics in DARWIN with a semi-supervised deep learning pipeline"*, J. Aalbers et al., Eur. Phys. J. C **86**, 312 (2026), arXiv:2410.00755 [physics.ins-det].

7. *"Towards the direct detection of composite ultraheavy dark matter in quantum sensor arrays"*, **D. W. P. Amaral**, E. Cai, A. J. Long, J. Qin, and C. D. Tunnell, JCAP **06**, 035 (2026), arXiv:2512.10124 [hep-ph].

8. *"Magnetic levitation as a new probe of non-Newtonian gravity"*, **D. W. P. Amaral**, T. M. Fuchs, H. Ulbricht, and C. D. Tunnell, Phys. Rev. D **113**, L021101 (2026), arXiv:2506.17385 [hep-ph].

9. *"Constraining dark photon dark matter with radio silence from soliton mergers around supermassive black holes"*, **D. W. P. Amaral**, E. D. Schiappacasse, and H.-Y. Zhang, JCAP **07**, 104 (2026), arXiv:2509.08932 [hep-ph].

10. *"Fast Bayesian inference for neutrino non-standard interactions at dark matter direct detection experiments"*, **D. W. P. Amaral**, S. Liang, J. Qin, and C. Tunnell, Mach. Learn. Sci. Tech. **6**, 015049 (2025), arXiv:2405.14932 [cs.LG].

11. *"First Search for Ultralight Dark Matter Using a Magnetically Levitated Particle"*, **D. W. P. Amaral**, D. G. Uitenbroek, T. H. Oosterkamp, and C. D. Tunnell, Phys. Rev. Lett. **134**, 251001 (2025), arXiv:2409.03814 [hep-ph].

12. *"Mechanical sensors for ultraheavy dark matter searches via long-range forces"*, J. Qin, **D. W. P. Amaral** et al., Phys. Rev. D **112**, 072003 (2025), arXiv:2503.11645 [hep-ph].

13. *"Vector wave dark matter and terrestrial quantum sensors"*, **D. W. P. Amaral**, M. Jain, M. A. Amin, and C. Tunnell, JCAP **06**, 050 (2024), arXiv:2403.02381 [hep-ph].

14. *"Rescuing the primordial black holes all-dark matter hypothesis from the fast radio bursts tension"*, **D. W. P. Amaral** and E. D. Schiappacasse, Phys. Rev. D **110**, 083532 (2024), arXiv:2312.09285 [hep-ph].

15. *"Measuring the sterile neutrino mass in spallation source and direct detection experiments"*, D. Alonso-González, **D. W. P. Amaral**, A. Bariego-Quintana, D. Cerdeño, and M. de los Rios, JHEP **12**, 096 (2023), arXiv:2307.05176 [hep-ph].

16. *"A direct detection view of the neutrino NSI landscape"*, **D. W. P. Amaral**, D. Cerdeño, A. Cheek, and P. Foldenauer, JHEP **07**, 071 (2023), arXiv:2302.12846 [hep-ph].

17. *"Confirming $U(1)_{L_\mu - L_\tau}$ as a solution for $(g-2)_\mu$ with neutrinos"*, **D. W. P. Amaral**, D. G. Cerdeño, A. Cheek, and P. Foldenauer, Eur. Phys. J. C **81**, 861 (2021), arXiv:2104.03297.

18. *"Solar neutrino probes of the muon anomalous magnetic moment in the gauged* $\mathrm{U}(1)_{L_\mu - L_\tau}$*"*, **D. W. P. Amaral**, D. G. Cerdeño, P. Foldenauer, and E. Reid, JHEP **12**, 155 (2020), arXiv:2006.11225 [hep-ph].

CONFERENCE TALKS AND SEMINARS

Colloquia

1. **University of Minnesota, Duluth**, *"Exploring Physics Beyond the Standard Model with Solar Neutrinos and Dark Matter Detectors"* (Aug. 2022).

Invited Seminars

2. **Quantum Seminar, JGU, Germany**, *"First Search for Ultraheavy Dark Matter Using a Magnetically Levitated Particle"* (Nov. 2026).
3. **CEICO Seminar, FZU, Czech Republic**, *"Floating into New Physics with Levitated Ferromagnets"* (May 2026).
4. **ICTP HECAP Seminar, Trieste, Italy**, *"Opening a New Window on Dark Matter with Magnetic Levitation"* (May 2026).
5. **Fermilab CPC Seminar, Chicago, USA**, *"Levitated Sensors at the Frontier of Fundamental Physics"* (Jan. 2026).
6. **KICP Group Meeting Talk, University of Chicago, USA**, *"Ultralight Vector Dark Matter"* (Jan. 2026).
7. **Helmholtz Institute, Mainz, Germany**, *"Building the Case for Magnetic Levitation in Fundamental Physics: Dark Matter and Non-Newtonian Gravity"* (Nov. 2025).

8. **Leiden University, The Netherlands**, *"Building the Case for Magnetic Levitation in Fundamental Physics: Dark Matter and Non-Newtonian Gravity"* (Nov. 2025).
9. **IFAE, Barcelona, Spain**, *"Magnetic Levitation for New Physics: From Dark Matter to Gravity"* (Nov. 2025).
10. **IPPP, Durham University, UK**, *"Hunting for Dark Matter Using Optomechanical Sensors"* (Jan. 2025).
11. **Rice University, Texas**, *"First Search for Ultralight Dark Matter Using a Magnetically Levitated Particle"* (Sept. 2024).
12. **Mitchell Institute, Texas A&M**, *"A Direct Detection View of the Neutrino NSI Landscape"*, video (Mar. 2023).
13. **Rice University, Texas**, *"Discovering New Neutrino Physics in Dark Matter Direct Detection Experiments with Solar Neutrinos"*, video (Mar. 2022).

Invited Conference and Workshop Talks

14. **Listening to the Cosmos: New Frontiers in Gravitational Wave Physics Workshop, GGI, Italy**, *"Searching for Primordial Black Hole Mergers with Electromagnetic Cavities"* (Sept. 2026).
15. **Mitchell Conference 2025, Texas A & M**, *"Searching for Dark Matter and Fifth Forces Using Magnetic Levitation"* (May 2025).
16. **TACOS 2023, Rice University**, *"Surfing Dark Matter Waves at the Windchime Experiment"*, slides (Oct. 2023).
17. **Magnificent CEνNS, Munich**, *"A Direct Detection View of the Neutrino NSI Landscape"*, video (Mar. 2023).
18. **MultiDark 2022, Madrid**, *"Probing Neutrino Non-Standard Interactions in Dark Matter Direct Detection Experiments"*, slides (May 2022).
19. **UK HEP Forum 2021**, *"Catching a Criminal: Confirming $U(1)_{L_\mu - L_\tau}$ as the Solution to $(g-2)_\mu$ with Neutrinos"*, slides (Nov. 2021).
20. **WIN2021**, *"Exploring the $U(1)_{L_\mu - L_\tau}$ Solution to the Muon's Anomalous Magnetic Moment Using Future Experimental Probes"*, slides (July 2021).

Parallel Conference Talks

21. **PLANCK 2026, CERN, Switzerland**, *"Searching for Ultraheavy Dark Matter with Magnetic Levitation"*, slides (May 2026).
22. **TeVPA 2025, Valencia**, *"Floating into the Unknown: Levitated Sensor for New Physics"*, slides (Nov. 2025).
23. **Pheno 2025, Pittsburgh**, *"Magnetic Levitation for Fundamental Physics: From Dark Matter to Non-Newtonian Gravity"* (May 2025).
24. **TAUP 2023, Vienna**, *"Riding Dark Matter Waves at Windchime"*, slides (Aug. 2023).

25. **SUSY 2023, Southampton**, *"Towards the Detection of Ultralight Dark Photon Dark Matter Using Optomechanical Sensors"*, slides (July 2023).
26. **Pheno 2023, Pittsburgh**, *"Detecting Ultralight Dark Photon Dark Matter Using Optomechanical Sensors"*, slides (May 2023).
27. **Texas Section of the APS, Rice University**, *"Constraining Neutrino Non-Standard Interactions in Dark Matter Direct Detection Experiments"* (Oct. 2022).
28. **YTF 20**, *"Machine Learning Surrogates for Rapid Dark Matter Direct Detection Calculations"*, slides (Dec. 2020).
29. **BUSSTEPP 2019**, *"Using Machine Learning and AI Techniques for the Direct Detection of Dark Matter at SuperCDMS"*, PhD forum talk (Aug. 2019).

SUPERVISING & TEACHING EXPERIENCE

Supervision

2026 **Martí Abella Cortes & Fatih Žgalj** (Undergraduates, IFAE)
Co-supervised summer students on project exploring axion-matter effects.

2025–2026 **Ferran Serra Vancells** (Master's Student, IFAE)
Supervised Master's thesis project on detecting gravitational waves using magnetic levitation experiments.
Thesis Title: *"Detecting Gravitational Waves with Magnetic Levitation"*.

2024–Present **Yitang Chen** (Undergraduate, Rice University)
Supervising projects on magnetic levitation for new physics searches and ultralight dark matter. Currently working on a publication in the manuscript stage.

2022–2025 **Erqian Cai** (Undergraduate, Rice University)
Supervised a summer research project and co-supervised a Bachelor's thesis with Andrew Long on quantum sensing for dark matter detection, resulting in the publication of *"Towards the direct detection of composite ultraheavy dark matter in quantum sensor arrays"*, arXiv:2512.10124 [hep-ph].

2019–2021 **Tutor, Durham University**
Level 1 Foundations of Physics
Personal tutor for two groups of 6 students each year, leading tutorials and marking assignments and examinations.

Teaching

2025 **Statistics for Physicists Lecturer, Rice University**
Delivered part of a statistics for physicists lecture course, teaching hypothesis testing in a frequentist setting.

2024 **QFT Guest Lecturer, Rice University**
Delivered part of a Quantum Field Theory lecture course.

2018–2019 **Demonstrator, Durham University**
Level 2 Mathematical Methods in Physics
Demonstrated problem-solving techniques alongside course lecturer, marking assignments and examinations.

2018 **Assessment Developer, Durham University**
Level 2 Laboratory Skills and Electronics
IPPP Summer Internship: Developed course material as Jupyter notebooks
Supervisor: Dr D. Maître.

2017 **Course Materials Developer, Durham University**
Level 3 Foundations of Physics 3A
IPPP Summer Internship: Designed a series of interactive Jupyter notebooks demonstrating core nuclear and particle physics concepts.
Supervisor: Dr D. Maître.

INDUSTRIAL RESEARCH EXPERIENCE

2021 **Geoptic, Research & Development**
Developed machine learning tools for use in muography, including neural networks for classification tasks and variational autoencoders for anomaly detection.

2019 **IBEX Innovations, Research & Development**
Created a Python package for medical X-ray imagingfor contrast enhancement, detail enhancement, and noise reduction. Results used in a successful clinical trial.

SCIENTIFIC SERVICE

Journal Reviewer: Reviewed for PRD and EPJC.

Workshop Organiser: Organiser of workshop on ultra-high frequency gravitational waves in Barcelona, Spain.

Seminar Series Organiser, IFAE (2025–Present): Organising weekly seminar series featuring both internal and external speakers.

Graduate Mentor, IFAE (2025–Present): Providing academic and pastoral support for a graduate student.

Member of EDI Committee, IFAE, (2025–Present): Helping to improve EDI commitments at IFAE.

Journal Club Organiser, IFAE, (2025–2026): Led weekly journal club.

Texas Section of the APS 2023: Session chair and judge.

TACOS 23: Member of organising committee for the TACOS 23 workshop. Session chair.

YTF 21: Chair of organising committee. Led the organisation of the YTF 21 conference.

YTF 20: Member of organising committee for the YTF 20 conference.

OUTREACH

Mathematics Symposium Invited Talk, Ramsey Grammar School (2025): Gave an invited colloquium-style talk on my research to A-level students (ages 17–18) attending an Island-wide mathematics competition.

Career Development Invited Talk, University of Houston (2023): Gave an invited colloquium-style talk on my research to undergraduate and graduate students. Spoke about pursuing the academic career path.

Career Development Invited Talk, Rice University (2022): Spoke to graduate students about pursuing the academic career path.

Biosciences Diversity Panelist, Rice University (2022): Spoke to undergraduate and graduate students about equality and diversity in the natural sciences.

Nuffield Research Placement Supervisor, Durham University (2022): Co-wrote a successful research proposal to host two Sixth-form students for a summer placement. Co-supervised a research project on neutrino physics beyond the Standard Model, leading to research talks in the UK.

Durham University Open Day Representative (2019): Spoke to prospective students about the physics course and undergraduate experience at Durham University.

Schools' Science Festival (2019): Volunteer at regional school event. Spoke to students about how direct detection experiments search for dark matter.

AWARDS & HONOURS

Three Minute Thesis Competition, Finalist (2022)
Finalist at public speaking competition for communicating thesis research.

Durham Excellence in Learning and Teaching Award (2021)
Led to a successful application to the Higher Education Academy for Associate Fellowship.

PROFESSIONAL AFFILIATIONS

2021–Present Associate Fellow of the Higher Education Academy (AFHEA)

Member of the **GravNet** and **Windchime** collaborations.

SKILLS

Computing: Python, Mathematica, C, C++

Languages: English (native), Portuguese (native), Italian (competent), French (basic), Spanish (basic)